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Abstract

Aim: To propose a method which analyses the electrocardiogram (ECG) waveform of any cardiac rhythm occurring during resuscitation and computes the probability of that rhythm converting into another with better prognosis (Pdes).

Methods: Rhythm transitions occurring spontaneously or due to defibrillation were analyzed. For each possible rhythm, ventricular fibrillation/ventricular tachycardia (VF/VT), pulseless electrical activity (PEA), pulse-generating rhythm (PR) and asystole (AS), the desired and undesired transitions were defined. ECG segments corresponding to the last 3s of rhythms prior to transition were used to extract waveform features. For each rhythm type, waveform features were combined into a logistic regression model to develop a rhythm specific classifier of desired transitions. This model was the monitoring function for the Pdes. The capacity of each rhythm specific classifier to discriminate between desired and undesired transitions was evaluated in terms of area under the curve (AUC). Pdes was integrated into a state sequence representation, which structures the information of cardiac arrest episodes, to analyze the effect of therapy on patient. As a case study, the effect of optimal/suboptimal cardiopulmonary resuscitation (CPR) on Pdes was analyzed. The mean Pdes was computed for the pre- and post-CPR intervals which presented the same underlying rhythm. The relationship between the optimal/suboptimal CPR and increase/decrease of Pdes was analyzed.

Results: The AUC was 0.80, 0.79, 0.73 and 0.61 for VF/VT, PEA, PR and AS respectively. The Pdes quantified the probability of every rhythm of the episode developing to a better state, and the evolution of Pdes was coherent with the provided therapy. The case study indicated, for most rhythms, that positive trends in the dynamic behaviour could be associated with optimal CPR, whereas the opposite seemed true for negative trends.

Conclusion: A method for continuous ECG waveform analysis covering all cardiac rhythms during resuscitation has been proposed. This methodology can be further developed to be used in retrospective studies of CPR techniques, and, in the future, for potentially monitoring in real time the probability of survival of patients being resuscitated.

Keywords: Cardiac arrest; ECG waveform analysis; Electrocardiogram (ECG); Rhythm transition classifier.

Beyond ventricular fibrillation analysis: comprehensive waveform analysis for all cardiac rhythms occurring during resuscitation

Erik Alonso*[,a,b], Trygve Eftestøl[a], Elisabete Aramendi[b], Jo Kramer-Johansen[c], Eirik Skogvoll[d,e], Trond Nordseth[d,e]

Affiliation and addresses:

[a] Department of Electrical Engineering and Computer Science.
University of Stavanger.
4036 Stavanger, Norway

[b] Communications Engineering Department.
University of the Basque Country UPV/EHU.
Alameda Urquijo S/N
48013 Bilbao, Spain

[c] National Advisory Unit for Prehospital Emergency Medicine (NAKOS).
Oslo University Hospital and University of Oslo.
N-0424 Oslo, Norway

[d] Institute for circulation and medical imaging.
Norwegian University of Science and Technology (NTNU).
N-7491 Trondheim, Norway

[e] Department of Anaesthesia and intensive care.
St. Olav University Hospital.
N-7014 Trondheim, Norway

Corresponding author:

* Erik Alonso

email: erik_alonso@ehu.es

Tel. : +34946017384

Fax. : +34946014259

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Abstract

Aim: To propose a method to analyse the waveforms for all cardiac rhythms occurring during resuscitation aiming to provide comprehensive analysis of the relationship between therapy and patient response.

Methods: Rhythm transitions occurring spontaneously or due to defibrillation were analysed. For each possible rhythm, ventricular fibrillation/ventricular tachycardia (VF/VT), pulseless electrical activity (PEA), pulse-generating rhythm (PR) and asystole (AS), the desired and undesired transitions were defined. ECG segments corresponding to the last 3 s of rhythms prior to transition were used to extract waveform features. For each rhythm type, waveform features were combined into a logistic regression model to develop a rhythm specific classifier of desired transitions. This model was the monitoring function for the probability of desired transition, P_{des}. Ten-fold cross validation was used to evaluate the discriminative power of the rhythm specific models in terms of area under the curve (AUC). P_{des} was integrated into a state sequence representation, which structures the information of cardiac arrest episodes, to analyse the effect of therapy on patient. As a case study, the effect of optimal/suboptimal cardiopulmonary resuscitation (CPR) on P_{des} was analysed. The mean P_{des} was computed for the pre- and post-CPR intervals which presented the same underlying rhythm. The relationship between the optimal/suboptimal CPR and increase/decrease of P_{des} was analysed.

Results: The AUC was 0.80, 0.79, 0.73 and 0.61 for VF/VT, PEA, PR and AS respectively. The P_{des} showed dynamic behaviour coherent with the provided therapy throughout the episode. The case study indicated, for most rhythms, that positive trends in the dynamic behaviour could be associated with optimal CPR, whereas the opposite seemed true for negative trends.

Conclusion: A method for continuous waveform analysis covering all cardiac rhythms during resuscitation has been proposed. This methodology can be further developed to explore optimal treatment strategies for individual patients during resuscitation.

Keywords

Cardiac Arrest, Electrocardiogram (ECG), ECG Waveform Analysis, Rhythm Transition Classifier

1. INTRODUCTION

In patients who suffer sudden cardiac arrest, rescuers provide ventilations, chest compressions, defibrillations and intravenous drugs in an effort to get the patient back to life. Patients may elicit different responses to this therapy, which can be observed from the cardiac rhythm changing dynamically between rhythms such as asystole (AS), ventricular fibrillation (VF), ventricular tachycardia (VT), pulseless electrical activity (PEA) and pulse-generating rhythms (PR), i.e. return of spontaneous circulation (ROSC). For patients in VF or VT the rescuer aims to abolish these rhythms by delivering an electrical shock. If the shock is successful, the heart recovers in PR. Clinical observations of VF waveforms on the defibrillators' display during resuscitation, made experienced rescuers able to associate the coarseness of the VF waveform with shock success. This inspired the computerised analysis of the VF waveform by calculating the mean amplitude of VF from consecutive waveform windows of a few seconds.[1]

Many research groups have developed VF waveform analysis techniques in time domain, frequency domain, wavelet domain and non linear techniques.[2–11] In human data, the VF waveform features were evaluated in terms of their ability to predict outcome; immediate defibrillation success being one of the most important indicators. VF waveform analysis was eventually recognised by the resuscitation research community, as can be seen in the Consensus on Science and cardiopulmonary resuscitation (CPR) guideline documents from both 2005 and 2010.[12,13]

There are two main applications of VF waveform analysis. On the one hand, some automated external defibrillators (AEDs), for instance the Philips FR2+ with smart CPR, offer a variant of VF waveform analysis to decide between immediate defibrillation or an initial period of chest compressions.[14] On the other hand, VF waveform analysis is used to study the relationship between therapy and patient response. Eftestøl et al. proposed a function to monitor the probability of successful defibrillation for retrospective analysis of the VF waveform.[15] An important aspect of this function is that it combines several defibrillation outcome predicting features into one single measure which can be used to monitor the changes in the VF waveform. This function was used to study the effect of interrupting chest compressions,[16] and the effect of varying durations of chest compression sequences.[17] Also, other features have been used to investigate how CPR quality influences the VF waveform.[18]

Recent studies have focussed on state sequence analysis of rhythm transitions during cardiac arrest.[19–21] The analysis of the prevalence and factors influencing transitions provide valuable information to define optimal loops and treatment during resuscitation.[22] It has been reported that the VF is not the most prevalent initial rhythm,[23] and when the effect of the therapy on whole episodes has been investigated,[18,20,24] it has become evident that the VF waveform analysis offers only a fragmented view of patient response during therapy. Nevertheless, to our knowledge, there is no method proposed to analyse the waveform during non-VF rhythms. Our objective in the current study is therefore to propose a method to analyse the waveforms for all rhythms occurring during resuscitation. This would allow the integration of state sequence and waveform analysis to provide comprehensive analysis of the relationship between therapy and patient response. To achieve this, rhythm specific classifiers will be proposed which quantify the electrocardiogram (ECG) waveform characteristics and relate them to the probability of patient improvement. Such classifiers are designed by collecting waveform features in form of measurements from waveforms immediately prior to rhythm transition. These measurements are categorised into two classes: desired (class 1) corresponding to ECG segments which convert into a rhythm with better prognosis, and undesired (class 2) which evolve towards a less desirable rhythm. The probability of getting the desired rhythm transition, P_{des}, will quantify the probability of any rhythm observed during resuscitation to convert to a more desired rhythm.

2. MATERIALS AND METHODS

In this section, the classifier that permits to discriminate the characterisation of the waveform into the two classes will be designed. Firstly, the P_{des} and the desired(class 1)/undesired(class 2) transitions will be defined for each rhythm category. Secondly, P_{des} will be shown integrated into a state sequence representation of out-of-hospital cardiac arrest (OHCA) episodes and evaluated as a CPR quality indicator.

2.1. Data materials

The data were obtained from a prospective study of 304 OHCA episodes, recorded between March 2002 and September 2004 in Akershus, Stockholm, and London.[25,26] The study was designed to measure the quality of out-of-hospital CPR performed by ambulance personnel in adherence to the resuscitation guidelines 2000.[27,28] HeartStart 4000 defibrillators (Philips Medical Systems, Andover, MA, USA) equipped with an additional chest pad were used to record with a sampling rate of 500 Hz the surface ECG (resolution 1.03 μV per least significant bit with 0-50 Hz bandwidth), the thoracic impedance, and additional reference channels used to compute the compression depth (CD) signal. Expert reviewers annotated the episodes as VF or VT in the shockable category, and as AS, PEA or PR in the non-shockable category.

To design the classifier, rhythm transitions occurring spontaneously or due to defibrillation were used. ECG segments of an initial rhythm (r_1) with a duration of 3 s were extracted immediately prior to transitions to a new rhtyhm (r_2). Segments of each class were associated to desired (class 1) or undesired (class 2) transitions. VF and VT were treated as a single rhythm category for which class 1 segments were those followed by PR after defibrillation. The class 1 for AS segments were those with transition to PR or PEA. The former is always desired, although it is less likely to spontaneously occur from AS, and the latter is an organized rhythm which could have an immediate way to PR. On the other hand, the class 2 for AS were those that convert into VF/VT which is undesirable as the myocardial oxygen consumption is higher than for PEA,[29,30] and the patient has no way of getting PR unless VF/VT is abolished. For PEA rhythm category, segments with transition to PR are considered as class 1. ROSC segments were considered as class 1 when the rhythm was maintained at least 10 s and corresponded to the episode ending rhythm.

As a case study, we investigated the effect of CPR on the P_{des}. ECG segments containing three time intervals with the following pattern were selected: $r_1 - Cr_1 - r_1$, where r_1 was any rhythm occurring during resuscitation, and Cr_1 corresponds to the interval when the r_1 rhythm is corrupted by CPR artifact. Minimum duration of 3 s was required for each interval of the pattern.

2.2. The waveform classifier

The waveform classifier was designed on the basis of the waveform features of the ECG signal segment, ecg[n], and the definition of a rhythm specific P_{des}.

Seven waveform features (v_i) were computed: mean slope, amplitude spectrum area (AMSA), centroid frequency, centroid power, mean amplitude, mean QRS amplitude and standard deviation of the QRS complex width. The features are well-known in the literature and a detailed description of their computation is provided in Appendix A.

A multivariate logistic regression model was applied to a rhythm specific combination of v_i features to compute P_{des} for each rhythm type (Appendix B). The discriminative power of the model was evaluated in terms of AUC (Area Under the Curve) using the Waikato Environment for Knowledge Analysis (WEKA) software.

2.3. State sequence representation

State sequence representation as proposed by Eftestøl et al.[31] is an effective method to structure information of OHCA episodes and give a compact visualisation at one glimpse. The integration of P_{des} in this representation permits comprehensive monitoring of the effect of therapy on the patient.

In the state sequence representation, criteria are defined to extract the ECG segments and to symbolically represent every rhythm.[31]

The P_{des} was computed sliding every second the 3 s analysis window of ecg[n]. Each symbol was then graphically represented by a filled and coloured polygon with variable amplitude between 0 and 1 according to the P_{des} value. The colour codes applied were VF/VT = red, PEA = yellow, ROSC = green, AS = grey. CPR corrupted intervals were represented by more intense colours and a horizontal black bar below the filled polygon. A black vertical line depicted a defibrillation. Rhythms with ongoing CPR were filtered using an adaptive scheme that incorporates a least mean square (LMS) algorithm[32] that estimates a noise-free ECG signal from which features can be extracted and P_{des} computed.

2.4. CPR effect on P_{des}: a case study

To analyse the effect of CPR on P_{des}, intervals with $r_1 - Cr_1 - r_1$ sequence were selected. The mean of the 3 s-P_{des} values was computed for the pre-CPR interval, P_{des_1}, and for the post-CPR interval, P_{des_2}. Difference was computed as:

$$\Delta P_{\text{des}} = P_{\text{des}_2} - P_{\text{des}_1} \tag{1}$$

Sequences with $\Delta P_{\text{des}} > 0.1$ were considered positively affected by CPR, while those with $\Delta P_{\text{des}} < -0.1$ were considered negatively affected.

The CPR of every interval was characterized in terms of mean compression depth (MCD), mean compression rate (MCR), percentage of compression with incomplete release ($PCIR$) and duration (d) of the CPR. These parameters were computed automatically using the CD signal in the CPR interval. These intervals were grouped into optimal and suboptimal classes. Intervals meeting the following criteria: $MCD \geq 38\,\text{mm}$, $MCR \geq 100\,\text{cpm}$, $PCIR \leq 10\,\%$ and $d \geq 18\,\text{s}$ were considered as optimal. By contrast, intervals with $MCD < 38\,\text{mm}$, $MCR < 100\,\text{cpm}$, $PCIR > 10\,\%$ and $d < 18\,\text{s}$ were considered as suboptimal. Relationship between $\Delta P_{\text{des}} > 0.1/\Delta P_{\text{des}} \leq -0.1$ and the optimal/suboptimal classes respectively was analysed.

3. RESULTS

Table 1 shows the number of segments per rhythm category analysed in the study, and the number of segments of each class associated to desired (class 1) or undesired (class 2) transitions.

Fig. 1 shows several segments of the database. From top to bottom, the first two panels correspond to a class 1 AS (which converts into PEA) and a class 2 PEA (which converts into VF/VT). The third panel shows a class 2 VF/VT, which converts into PEA due to defibrillation. Class 1 and class 2 ROSC segments can be seen in the last two panels, the first one did not last until the end of the episode, while the second one was the final rhythm of the episode. The results obtained for the classifier in terms of AUC were 0.80, 0.79, 0.73 and 0.61 for VF/VT, PEA, ROSC and AS respectively.

Fig. 2 and 3 show two examples of the integration of P_{des} into the described state sequence representation for a successful and an unsuccessful resuscitation case, respectively. Figure 2 shows how P_{des} for the VF increases with time until a shock is delivered when P_{des} is close to 1 (minute 11) and ROSC is achieved. By contrast, in Fig. 3 the P_{des} for the PEA remains low along the episode, and is close to 0.5 for the VF when shock is delivered (around minute 15). The unsuccessful shock prolonged the PEA with low P_{des} till the end of the episode.

Fig. 4 summarizes the results obtained for the case study. It represents the percentage of optimal(left)/suboptimal(right) rhythm sequences that showed an increase(left)/decrease(right) in P_{des} for each rhythm category. VF/VT, PEA, ROSC and AS are represented by red, yellow, green and grey bars respectively. The number of optimal/suboptimal rhythm sequences used in this case study for each different underlying rhythm can be observed below each bar. For instance, for the 56 sequences with VF as underlying rhythm (VF-CVF-VF) where the CPR provided was optimal, 47 (84%) showed an increase in P_{des} ($\Delta P_{\mathrm{des}} > 0.1$). By contrast, for the 2 VF-CVF-VF sequences where the CPR provided was suboptimal, all of them (100%) showed a decrease in P_{des} ($\Delta P_{\mathrm{des}} < -0.1$).

4. DISCUSSION

We have presented a methodology for expanding the concept of VF waveform analysis to encompass all rhythms encountered during resuscitation. By defining the two categories of desired and undesired rhythm transitions for each type of rhythm, we have demonstrated how a function for monitoring the probability of desired transition can be conceived.

We have shown how a complete resuscitation episode can be graphically visualized where transition states can be followed by colour coding, and the continuous indication of P_{des} shows the interplay between therapy and patient response. For example, in Fig. 2, one can see how the compression sequences influences the probability positively during VF from minutes 1 to 6. An immediate decrease prior to the shocks can also be seen, but the trend is overall positive, and eventually ROSC occurs at about minute 7. CPR during AS does not give a readily interpretable results, but there is eventually a spontaneous transition to ROSC following CPR. A positive effect of CPR is also indicated during PEA before the transition into PR at 12 minutes. Fig. 3 mainly shows changes in PEA due to therapy.

Finally, we have demonstrated how this way of representing the resuscitation data can be used as a framework for analysing the relationship between therapy and patient response. This was done by distinguishing between CPR sequences giving positive and negative changes in the probability responses, i.e. increase/decrease in P_{des}, for each rhythm type and comparing to the quality indication of the therapy. The results (Fig. 4) indicate that the positive and negative responses in probability are associated with optimal and suboptimal performance of therapy respectively in all the rhythm categories, except AS where results are inconclusive.

The last decade has witnessed a change in the prevalence of initial rhythms with the reduction in the proportion of VF/VT and a corresponding increase in the incidence of PEA and AS.[33] This situation calls for new strategies with increased focus on these rhythms.[34] Where the success or failure of shock therapy has been a focus for the therapy of VF/VT, the focus shifts to supplementary interventions to handle the underlying causes for the other rhythms.[34] Chest compressions for increased myocardial blood flow are emphasized as one of the key strategies, and thus the ability to monitor the effect of this therapy is of major importance to guide therapy in an optimal way.

Strategies for handling PEA has been perceived as being a factor of major importance for improved therapy.[34] PEA has many subgroups[34] and being able to distinguish between them might be a key to distinguish between cardiac states with different prognosis. It is our hypothesis that different subgroups can be associated with different transition rhythms and different P_{des}.

Our definitions of desired and undesired transitions are open for discussion. For example, we have defined VF/VT as the undesired transition from AS. There are two reasons for VF/VT being undesired: (1) the oxygen consumption is higher than for PEA/AS and since it is not an organised rhythm the patient has no immediate way to ROSC. (2) Most patients shocked from VF to PEA must get precordial compressions which might cause a new VF/VT. Therefore, we consider it better to be in PEA than in VF. For patients found in AS, the development of VF in the ECG signal may implicate that a very fine VF has developed into a coarse VF, which may be abolished by defibrillation and turned into PR if the shock is successful. So, in this case one might consider going to VF/VT as desirable. The uncertainty in the definition of desired rhythm from AS is reflected in the predictive power of P_{des} (AUC = 0.61).

There are limitations to this study. For the generation of the state sequence representations shown in Fig. 2 and 3, the probabilities were calculated directly from the filtered signals in the intervals corrupted by CPR. The filtering method applied was proven to maintain the predictive power of some of the features.[35] Although, we might expect possible residues of the filtering to affect the other features and possibly explain sudden changes in the probabilities around these transitions.

In the case study, the number of segments analysed in some cases are very few, so it is difficult to make any strong conclusions from this experiment which must be considered as preliminary.

5. Conclusions

A method for applying continuous waveform analysis covering all the cardiac rhythms during resuscitation has been proposed. The probability functions associated with the transitions of desire, show dynamic behaviour coherent with the evolution of the episode. A preliminary case study indicated for most rhythms that strong positive trends in the dynamic behaviour could be associated with CPR considered optimal according to the resuscitation guidelines, while the opposite seemed true for negative trends.

Conflict of interest

Author, E. Aramendi has received research support from Bexen Cardio (Ermua, Spain) for studies on AED shock advice algorithms.

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Appendix

A. Signal processing and feature extraction

ECG segments, $ecg[\mathrm{n}]$, corresponding to the last 3 s of the rhythm r_1 were processed to compute P_{des}. $ecg[\mathrm{n}]$ signal was first downsampled to 250 Hz and then, band-pass filtered between 0.7–30 Hz using an order 8 Butterworth filter in order to suppress baseline drift and high frequency noise. Seven different waveform features were computed as follows:

- **Mean slope, v_1:** Mean value of the sum of the first difference of $ecg[\mathrm{n}]$.

$$v_1(\frac{\mathrm{mV}}{\mathrm{s}}) = \frac{\sum_{n=1}^{N} |ecg[\mathrm{n}] - ecg[\mathrm{n\text{-}1}]|}{N} \tag{2}$$

  where N is the length in samples of $ecg[\mathrm{n}]$.

- **AMSA (AMplitude Spectrum Area), v_2:** Sum of the individual amplitudes and frequencies of the spectrum of $ecg[\mathrm{n}]$. The spectrum was computed using the FFT (Fast Fourier Transform) with 1024 points.

$$v_2(\mathrm{mV} \cdot \mathrm{Hz}) = \sum_{n=1}^{N_f} A_i \cdot f_i \tag{3}$$

where A_i represents the amplitude of the frequency component, f_i, for $i = 1..N_f$, being N_f the highest harmonic considered.

- **Centroid frequency, v_3:** Frequency bisecting the area under the power spectral density (PSD) which was computed using 1024 points Welch's averaged[36] for each 1 s segment windowed with a Hamming window.

$$v_3(\mathrm{Hz}) = \frac{\sum_{i=1}^{N_f} \mathrm{PSD}(f_i) \cdot f_i}{\sum_{i=1}^{N_f} \mathrm{PSD}(f_i)} \tag{4}$$

- **Centroid power, v_4:** Sum of the squared values of the PSD divided by sum of the values of the PSD.

$$v_4(\mathrm{mV}^2) = \frac{\sum_{i=1}^{N_f} PSD^2(f_i)}{\sum_{i=1}^{N_f} PSD(f_i)} \tag{5}$$

- **Mean amplitude, v_5:** Mean of the absolute value of $ecg[\mathrm{n}]$.

$$v_5(\mathrm{mV}) = \frac{\sum_{n=1}^{N} |ecg[\mathrm{n}]|}{N} \tag{6}$$

- **Mean QRS amplitude, v_6:** Mean of the absolute value of the QRS complex amplitude of $ecg[\mathrm{n}]$.

$$v_6(\mathrm{mV}) = \frac{\sum_{n=1}^{N_{\mathrm{QRS}}} |ecg[x_{\mathrm{n}}]|}{N_{\mathrm{QRS}}} \tag{7}$$

being N_{QRS} the number of QRS complexes within the $ecg[\mathrm{n}]$, and x_{n} the n^{th} QRS complex position in time samples.

- **Standard deviation of the QRS complex width, v_7:** Standard deviation of the QRS complexes width of $ecg[\mathrm{n}]$.

$$v_7(\mathrm{s}) = \sqrt{\frac{\sum_{n=1}^{N_{\mathrm{QRS}}} (w_{\mathrm{n}} - \mu)^2}{N_{\mathrm{QRS}}}} \tag{8}$$

where w_{n} denotes the n^{th} QRS width in seconds and μ refers to the mean width of the N_{QRS} QRS complexes. The width of the QRS complex was defined as follows:

$$w_n(s) = on_{\mathrm{n}} - off_{\mathrm{n}} \tag{9}$$

where $on_{\mathrm{n}}/off_{\mathrm{n}}$ are the onset/offset instants of the n^{th} QRS complex respectively. They were computed as the last/first zero-crossings in the first difference of ecg[n] before/after x_{n} respectively.

B. Probability of the desired transition, P_{des}

A multivariate logistic repression model was applied to a rhythm specific combination of ECG features (v_i) to compute P_{des}. The model assigns to each ecg[n] a P_{des} given by:

$$P_{\mathrm{des}}(z) = \frac{1}{1 + e^{-z}} \tag{10}$$

with $z = \theta_0 + \theta_1 \cdot v_1 + \theta_2 \cdot v_2 + \theta_3 \cdot v_3 + \theta_4 \cdot v_4 + \theta_5 \cdot v_5 + \theta_6 \cdot v_6 + \theta_7 \cdot v_7$

The rhythm specific features used to build the multivariate logistic regression model are shown in Table 2. Ten-fold cross validation was used to calculate the regression coefficients (θ_i). ecg[n] segments with an assigned $P_{\mathrm{des}} > 0.5$ were classified as class 1. Imbalance between classes was compensated weighting each sample according to the proportion of samples in each class.

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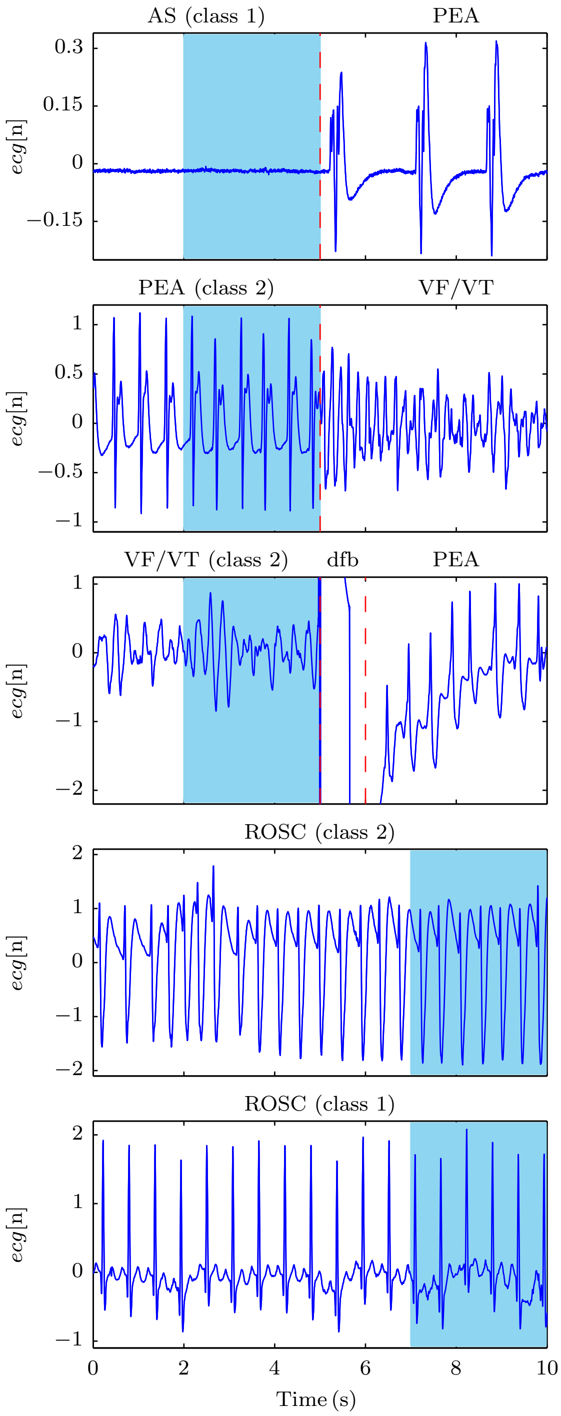
Figure Legends

Figure 1	Examples of different rhythm transitions. From top to bottom, a desired (class 1) AS which converts into PEA, a class 2 PEA with transition to to VF/VT, a class 2 VF/VT with transition to PEA caused by an electrical shock (dfb), and class 2 and class 1 ROSC examples respectively. The 3 s ECG segments used to develop the classifier are coloured in blue.
Figure 2	State sequence representation example for a successful resuscitation episode. Time periods where patients received chest compressions are shaded in the corresponding colour of the current state and represented with a horizontal black line below the filled polygon. The black vertical line depicts a defibrillation.
Figure 3	State sequence representation example for an unsuccessful resuscitation episode. Time periods where patients received chest compressions are shaded in the corresponding colour of the current state and represented with a horizontal black line below the filled polygon. The black vertical line depicts a defibrillation.
Figure 4	Summary of the results obtained from the case study. The percentage of optimal(left)/suboptimal(right) CPR sequences that showed an increase (left)/decrease (right) in P_{des} for each rhythm category. VF/VT, PEA, ROSC and AS are represented by red, yellow, green and grey bars respectively. The number of optimal/suboptimal rhythm sequences used in this case study for each different underlying rhythm can be observed below each bar.



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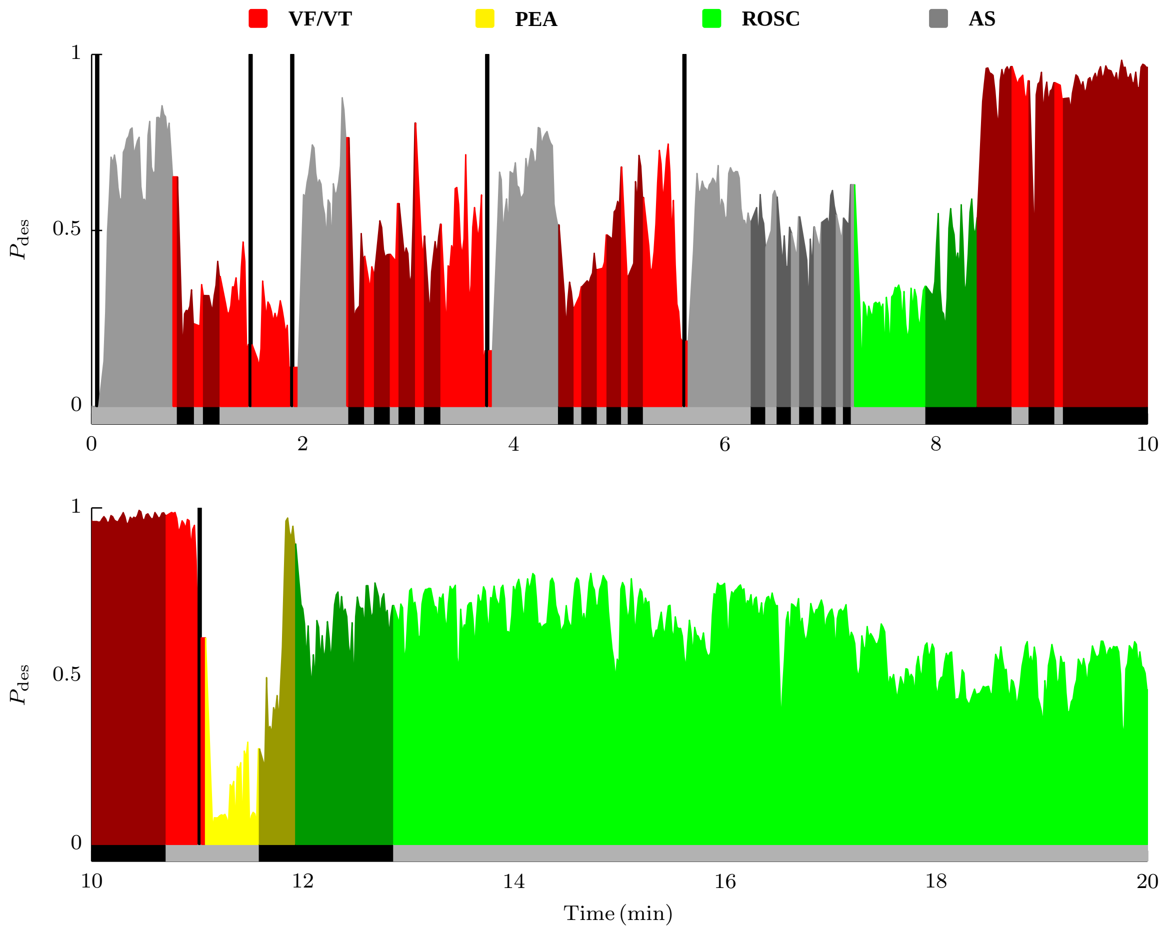

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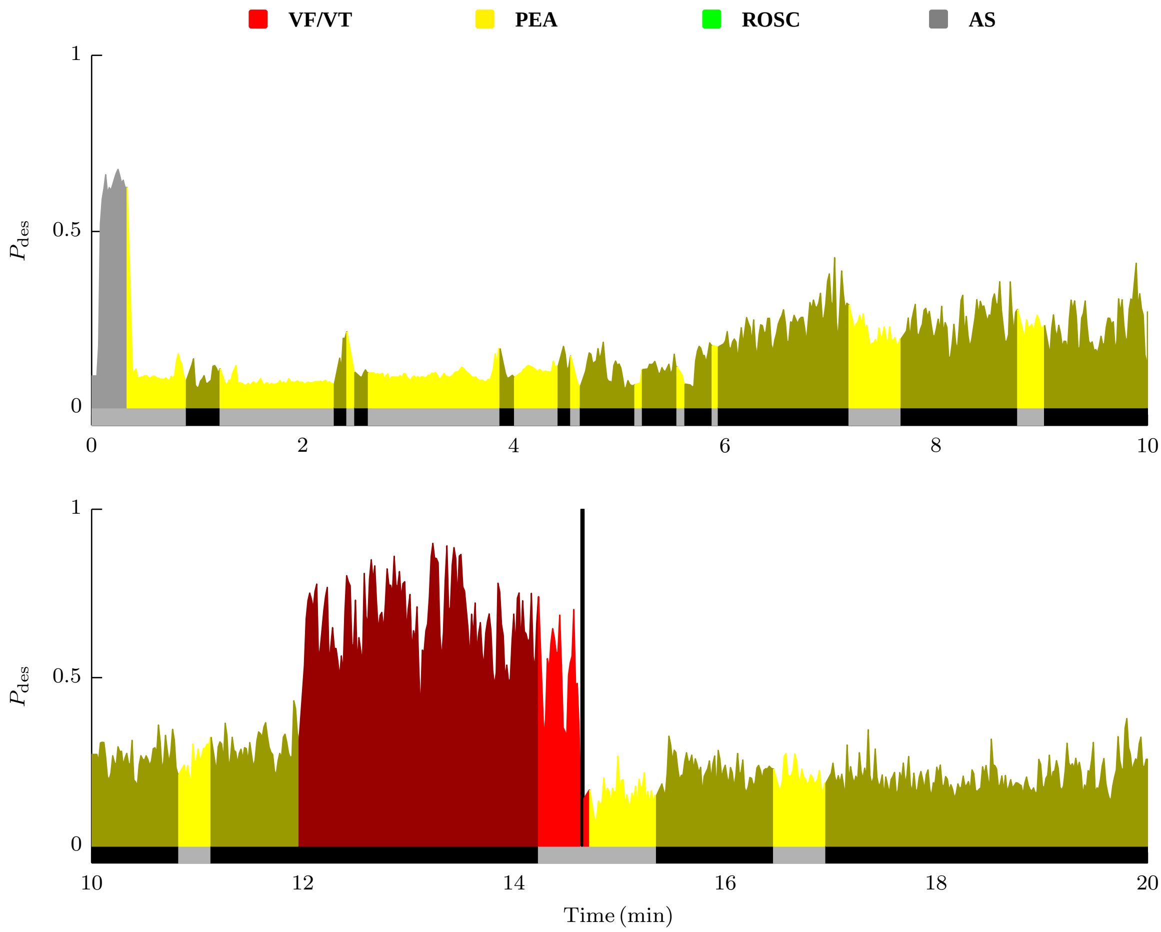

Figure 3: State sequence representation example for an unsuccessful resuscitation episode. Time periods where patients received chest compressions are shaded in the corresponding colour of the current state and represented with a horizontal black line below the filled polygon. The black vertical line depicts a defibrillation.

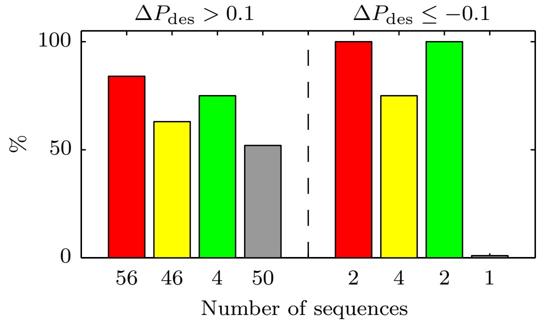

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Table Legends

Table 1	Number of segments associated to desired (class 1) and undesired (class 2) classes for each rhythm category.
Table 2	Summary of the rhythm specific features, v_i, used to build the multivariate logistic regression model.

Table 1: Number of segments associated to desired (class 1) and undesired (class 2) classes for each rhythm category.

Rhythm Category	**Class 1 (desired)**	**Class 2 (undesired)**	**Total**
VF/VT	40	708	748
PEA	82	158	240
ROSC	66	456	522
AS	87	64	151

Table 2: Summary of the rhythm specific features, v_i, used to build the multivariate logistic regression model.

Rhythm Category	v_1	v_2	v_3	v_4	v_5	v_6	v_7
VF/VT			X		X		
PEA	X	X	X	X		X	X
ROSC			X		X		X
AS		X	X				